

SECTION 1

SOFTWARE REQUIREMENTS SPECIFICATION

SE Mini Project

TITLE: Student Record System

TEAM NO: 11

TEAM MEMBERS

INDIVIDUAL CONTRIBUTION

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Section 4 – System Features (Detailed)

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Section 6 – Quality Attributes & Acceptance Tests

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Section 2 – Overall Description

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Section 7 – System Models and Diagrams

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Section 8 – Requirements Traceability Matrix (RTM)

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Section 1 – Introduction

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Section 5 – Non-Functional Requirements (Detailed)

SECTION 1:

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a **Student Record System**. It defines functional and non-functional requirements, interfaces, and verification criteria for a system designed to efficiently manage student information including grades, attendance, and personal details. This document serves as a reference for developers, testers, and project stakeholders throughout the development lifecycle.

1.2 Scope

The Student Record System covers the management of student data including:

- Recording and updating student personal information (name, ID, contact details, etc.)
- Tracking and managing student grades across multiple subjects and semesters
- Recording and monitoring student attendance
- Generating reports such as grade sheets, attendance summaries, and student profiles
- Providing role-based access for administrators, faculty, authorized staff, and students

The system excludes integration with external learning management systems, financial management (fees, scholarships), and examination scheduling beyond grade entry.

1.3 Audience

- **Developers** – To understand system requirements and build the software
- **QA Engineers** – To design and execute test cases
- **System Administrators** – To deploy and maintain the system
- **End Users (Faculty, Administrators, Authorized Staff, Students)** – To understand system capabilities
- **Project Evaluators** – To assess the completeness of the system

1.4 Definitions and Acronyms

Term	Definition
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SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
RDBMS	Relational Database Management System
RBAC	Role-Based Access Control
GPA	Grade Point Average
CGPA	Cumulative Grade Point Average
OTP	One-Time Password
JWT	JSON Web Token
TLS	Transport Layer Security

SECTION 2

2. Overall Description :

2.1 Product Perspective

The **Student Record System (SRS)** is a computerized record-management system designed to store, manage, retrieve, and maintain student-related academic information in a structured manner. The system provides a centralized platform for managing student records such as personal details, courses or subjects, grades, attendance, and academic reports.

The system is intended to reduce manual record-keeping and provide authorized users with quick and organized access to student information. It consists of application software, a database for persistent storage, and user interfaces through which users can perform record-management operations.

The system operates as an independent academic record-management application and may be used by authorized administrative or academic staff. Student information entered into the system is stored in the database and can subsequently be searched, updated, or used for generating reports.

2.2 Major Product Functions

The Student Record System shall provide the following major functions:

- Student Management — Add, update, view, and manage student information.
- Course / Subject Management — Maintain information about courses or subjects associated with students.
- Grade Management — Record, update, and retrieve student grades or marks.
- Attendance Management — Record and maintain student attendance information.
- Record Search — Search and retrieve student records using relevant student details.
- Reporting — Generate student-related academic and attendance reports.
- User / Access Management — Control access to system functions based on authorized users.
- Data Management — Store, update, and maintain student records in a structured database.
- Record Administration — Allow authorized users to manage existing records and perform required administrative operations.
- System Exit — Provide a controlled mechanism for users to safely exit the system.

2.3 User Roles and Characteristics

The Student Record System shall support the following primary user roles:

1. Administrator :

The Administrator is responsible for managing the overall system and maintaining student-related data. The administrator may add, update, delete, and manage student records, courses, grades, attendance information, and authorized user access.

2. Faculty / Teacher :

Faculty members use the system to manage academic information related to students. They may record or update grades and attendance and retrieve student records required for academic activities.

3. Student :

Students may access their permitted academic information, such as personal records, course details, grades, and attendance, depending on the access permissions provided by the system.

4. Authorized Staff :

Administrative or academic staff may use the system to search, view, update, and generate student-related records according to their assigned permissions.

Users are expected to have basic computer knowledge and familiarity with student academic information. Administrative users may require additional knowledge of record management and system administration.

2.4 Operating Environment

The Student Record System is intended to operate in a standard computing environment consisting of:

- A desktop or laptop computer capable of running the application.
- A suitable operating system such as Windows or Linux.
- A database management system for storing student records.
- An application interface for user interaction.
- Sufficient storage for maintaining student records and related academic information.
- A local network or internet connection if the system is implemented as a network-based application.

2.5 Constraints

The Student Record System is subject to the following constraints:

- Only authorized users shall be permitted to access protected student information.
- The system shall maintain the consistency and integrity of student records.
- User access shall be restricted according to the user's assigned role.
- The database structure shall support reliable storage and retrieval of student information.
- The system shall operate within the hardware and software resources available in the intended environment.
- Student information shall be handled securely and shall not be unnecessarily exposed to unauthorized users.
- The implementation shall conform to the programming language, database, and development tools selected for the project.

SECTION 3

3. External Interface Requirements

3.1 User Interfaces

Primary UI: The Student Record System shall provide a clear and structured user interface through which users can perform operations related to student records, including viewing, adding, updating, searching, and managing student information, grades, and attendance.

Accessibility: The interface shall provide clear prompts, understandable messages, and simple navigation to help users perform the required operations effectively.

3.2 Hardware Interfaces

- **Keyboard** — used for entering student information, grades, attendance details, search criteria, and menu selections.
- **Display/Monitor** — used to display menus, student records, grades, attendance information, and system messages.
- **Mouse** — may be supported depending on the final user-interface implementation.
- **Standard computer system** — used to run the Student Record System.

3.3 Software Interfaces

- **C/C++ runtime environment** — used to execute the Student Record System.
- **Data storage interface** — used for storing and retrieving student records, grades, and attendance information.
- **Database Management System** — used to store, retrieve, update, and manage student, subject, grade, and attendance records.
- **No external third-party service integration** is required.

3.4 Communications

- The Student Record System shall operate as a standalone C/C++ application within the supported computing environment.
- The basic implementation shall not require network communication.
- Student records and other system data shall be transferred only through the interfaces supported by the application.

SECTION 4

4. System Features (Detailed)

4.1 Student Management

Description:

The Student Management feature allows authorized users to maintain student personal and academic information.

SRS-F-001: The system shall allow administrators to add new student records including student ID, name, contact details, and academic information.

Acceptance Criteria: A valid student record is successfully stored.

Test Case: TC-STD-01

SRS-F-002: The system shall allow authorized users to view student information.

Acceptance Criteria: The selected student's permitted information is displayed correctly.

Test Case: TC-STD-02

SRS-F-003: The system shall allow administrators to update existing student information.

Acceptance Criteria: Updated student information is saved and displayed correctly.

Test Case: TC-STD-03

SRS-F-004: The system shall prevent duplicate student IDs.

Acceptance Criteria: An attempt to create a duplicate student ID is rejected.

Test Case: TC-STD-04

4.2 Course / Subject Management

Description:

The Course / Subject Management feature allows administrators to maintain subjects and associate them with students.

SRS-F-005: The system shall allow administrators to add and maintain subject records.

Acceptance Criteria: A valid subject record is successfully created and stored.

Test Case: TC-SUB-01

SRS-F-006: The system shall allow authorized users to view available subjects.

Acceptance Criteria: Subject details are displayed correctly.

Test Case: TC-SUB-02

SRS-F-007: The system shall allow administrators to assign subjects to students.

Acceptance Criteria: The assigned subject appears in the student's academic record.

Test Case: TC-SUB-03

4.3 Grade Management

Description:

The Grade Management feature allows authorized faculty users to record, update, and view student grades.

SRS-F-008: The system shall allow authorized faculty users to enter student grades.

Acceptance Criteria: A valid grade is stored against the correct student and subject.

Test Case: TC-GRD-01

SRS-F-009: The system shall validate grades before storing them.

Acceptance Criteria: Invalid grade values are rejected.

Test Case: TC-GRD-02

SRS-F-010: The system shall allow authorized faculty users to update student grades.

Acceptance Criteria: The modified grade is correctly stored and displayed.

Test Case: TC-GRD-03

SRS-F-011: The system shall calculate GPA and CGPA from recorded grades.

Acceptance Criteria: The calculated GPA/CGPA matches the expected result.

Test Case: TC-GRD-04

4.4 Attendance Management

Description:

The Attendance Management feature allows authorized faculty users to record, update, and monitor student attendance.

SRS-F-012: The system shall allow authorized faculty users to record student attendance.

Acceptance Criteria: Attendance is recorded for the correct student and subject.

Test Case: TC-ATT-01

SRS-F-013: The system shall allow authorized faculty users to update attendance records.

Acceptance Criteria: Updated attendance information is correctly stored.

Test Case: TC-ATT-02

SRS-F-014: The system shall calculate attendance percentage for students.

Acceptance Criteria: The displayed percentage matches the recorded attendance.

Test Case: TC-ATT-03

4.5 Record Search and Reporting

Description:

The Record Search and Reporting feature allows authorized users to locate student records and generate academic reports.

SRS-F-015: The system shall allow authorized users to search for students using student ID or name.

Acceptance Criteria: Matching student records are displayed.

Test Case: TC-SRCH-01

SRS-F-016: The system shall generate student academic reports containing grades and attendance information.

Acceptance Criteria: The generated report contains the student's correct records.

Test Case: TC-RPT-01

SRS-F-017: The system shall allow users to view grade sheets and attendance summaries.

Acceptance Criteria: The selected grade sheet or attendance summary is displayed correctly.

Test Case: TC-RPT-02

4.6 User / Access Management

Description:

The User / Access Management feature provides authentication and role-based access for administrators, faculty, authorized staff, and students.

SRS-F-018: The system shall authenticate users before granting access.

Acceptance Criteria: Valid credentials provide access and invalid credentials are rejected.

Test Case: TC-AUTH-01

SRS-F-019: The system shall provide role-based access for administrators, faculty, authorized staff, and students.

Acceptance Criteria: Users can access only functions permitted for their role.

Test Case: TC-AUTH-02

SRS-F-020: The system shall prevent students from modifying protected academic records.

Acceptance Criteria: Unauthorized modification attempts are rejected.

Test Case: TC-AUTH-03

4.7 Data Management

Description:

The Data Management feature ensures that student, subject, grade, and attendance information is stored and maintained consistently.

SRS-F-021: The system shall store student, subject, grade, and attendance records in persistent storage.

Acceptance Criteria: Stored information remains available after restarting the system.

Test Case: TC-DATA-01

SRS-F-022: The system shall validate required information before storing records.

Acceptance Criteria: Records with missing required fields are rejected.

Test Case: TC-DATA-02

SRS-F-023: The system shall maintain consistency between students, subjects, grades, and attendance records.

Acceptance Criteria: Invalid references cannot be stored.

Test Case: TC-DATA-03

4.8 System Exit / Administration

Description:

The System Exit / Administration feature provides safe logout and administrative account-management functionality.

SRS-F-024: The system shall provide a logout or exit function.

Acceptance Criteria: Selecting logout or exit safely terminates the active session.

Test Case: TC-EXIT-01

SRS-F-025: The system shall allow administrators to manage authorized user accounts.

Acceptance Criteria: Administrators can create, modify, or deactivate permitted accounts.

Test Case: TC-ADMIN-01

SECTION 5

5. Non-Functional Requirements

NFRs below are measurable and tied to test plans. IDs follow SRS-NF-####.

5.1 Non-Functional Requirements Table

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
SRS-N F-001	The system shall load the student dashboard and list pages within 3 seconds for 95% of requests under normal load.	Performance	High	95th percentile response time $\leq$ 3s in load testing. Test: TC-Perf-01
SRS-N F-002	The system shall operate reliably during normal usage without unexpected termination or loss of stored records.	Reliability	High	The system completes normal operations without unexpected termination or data loss. Test: TC-REL-01
SRS-N F-003	The system shall protect stored student personal data from unauthorized access and shall not store sensitive information in plaintext where protection is required.	Security/Compliance	High	Encryption audit pass; no plaintext PII in logs or database. Test: TC-Sec-01
SRS-N F-004	The system shall maintain audit logs of all CRUD operations (who accessed/ modified what, and when) retained for 5 years .	Audit/Data Retention	High	Automated logging and archival verified. Test: TC-OPS-01

SRS-N F-005	The system shall support accessibility features (screen reader compatibility, keyboard navigation, high-contrast mode) conforming to WCAG 2.1 AA where applicable.	Usability/Accessibility	Medium	Accessibility audit pass. Test: TC-UX-01
SRS-N F-006	The system shall handle the expected number of student records efficiently without significant degradation in normal operation.	Scalability	Medium	The system successfully processes the defined test dataset without significant performance degradation during normal operation. Test: TC-Perf-02
SRS-N F-007	The system shall provide role-based access control (Admin, Faculty, Authorized Staff, Student) with distinct permissions for each role.	Security	High	RBAC matrix verified; unauthorized access attempts are blocked. Test: TC-Sec-02

5.2 Security Objectives

This section lists the security objectives for the Student Record System project.

Security Objective 1 (SO-001): Protect Student Data Confidentiality

All student personal information (names, IDs, contact details, grades, attendance records) must be protected from unauthorized access while stored. Only authorized users (faculty, administrators, authorized staff, and the student themselves) should have access to relevant data based on their role.

Security Objective 2 (SO-002): Ensure Data Integrity and Auditability

All records in the system (grades, attendance, personal details) must be protected from unauthorized modification. Every change must be logged with timestamp and user identity to enable complete audit trails and accountability.

5.3 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case Ref
SRS-S R-001	The system shall protect stored student information from unauthorized access .	Security	High	Unauthorized attempts to access protected student information are rejected. Test: TC-Sec-01
SRS-S R-002	All user authentication shall use secure password hashing (bcrypt/Argon2); passwords shall never be stored in plaintext.	Security	High	Password stored as hash; plaintext not found in logs/database. Test: TC-Sec-02
SRS-S R-003	The system shall enforce session timeout after 15 minutes of inactivity for all users.	Security	Medium	Session terminates at 15 minutes; user must re-authenticate. Test: TC-Sec-03
SRS-S R-004	The system shall log failed login attempts with a timestamp and restrict access after 5 consecutive failed attempts.	Security	High	Audit logs show failed attempts; account locks after 5 failures. Test: TC-Sec-04
SRS-S R-005	Role-based access control shall restrict user actions to only their permitted operations (Admin: full access; Faculty: manage grades/attendance; Authorized Staff: permitted record-management operations; Student: view own data).	Security	High	RBAC enforcement verified for all roles; unauthorized actions blocked. Test: TC-Sec-05

SECTION 6

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

Performance:

The system shall display normal student records and search results within 3 seconds under normal operating conditions.

Reliability:

The system shall maintain student, grade, attendance, and subject records accurately without unintended data loss during normal operation.

Usability:

The system shall provide clear menus, understandable messages, and simple navigation for administrators, faculty, and students.

Portability:

The system shall be capable of operating on the supported desktop operating environments specified for the project.

Maintainability:

The system shall use modular components so that student, subject, grade, attendance, and user-management functions can be maintained independently.

Security:

The system shall authenticate users and enforce role-based access to protect student personal and academic information.

Data Integrity:

The system shall validate student IDs, grades, attendance values, and subject references before storing information.

6.2 Acceptance / Exit Criteria

The Student Record System shall be considered acceptable when:

1. All functional requirements are implemented and verified.
2. Student records can be added, viewed, updated, and maintained correctly.
3. Subject records can be created, viewed, and assigned correctly.
4. Grades can be entered, validated, updated, and calculated correctly.
5. Attendance can be recorded, updated, and calculated correctly.
6. Student search and reporting functions operate correctly.
7. User authentication works correctly.
8. Role-based access restrictions work correctly.
9. Invalid input is appropriately rejected.
10. Stored records remain available after restarting the system.
11. No critical data-integrity failures remain.

12. All defined acceptance test cases have passed.
13. No unresolved critical defects remain before final submission.

6.3 Test Case Naming Convention

Test cases shall follow the format:

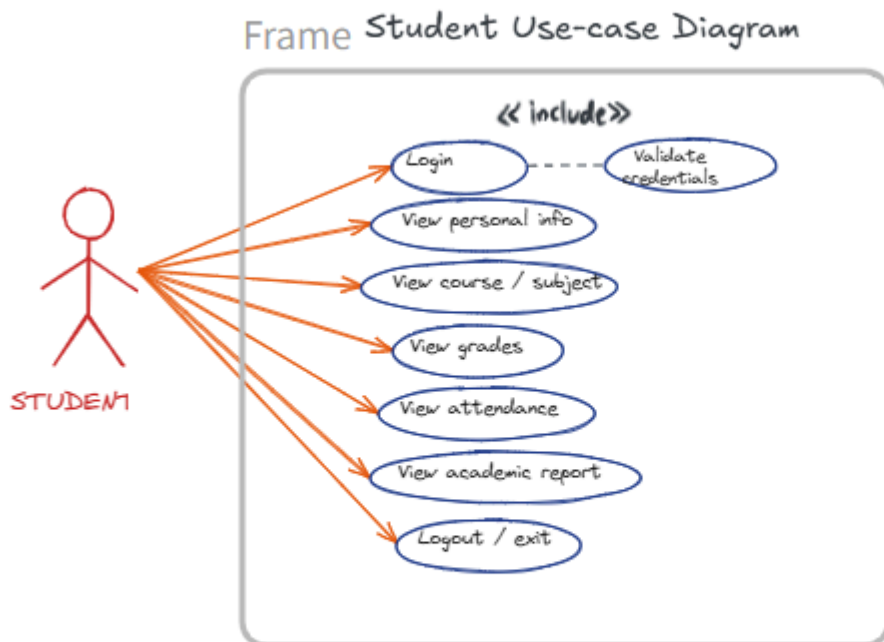
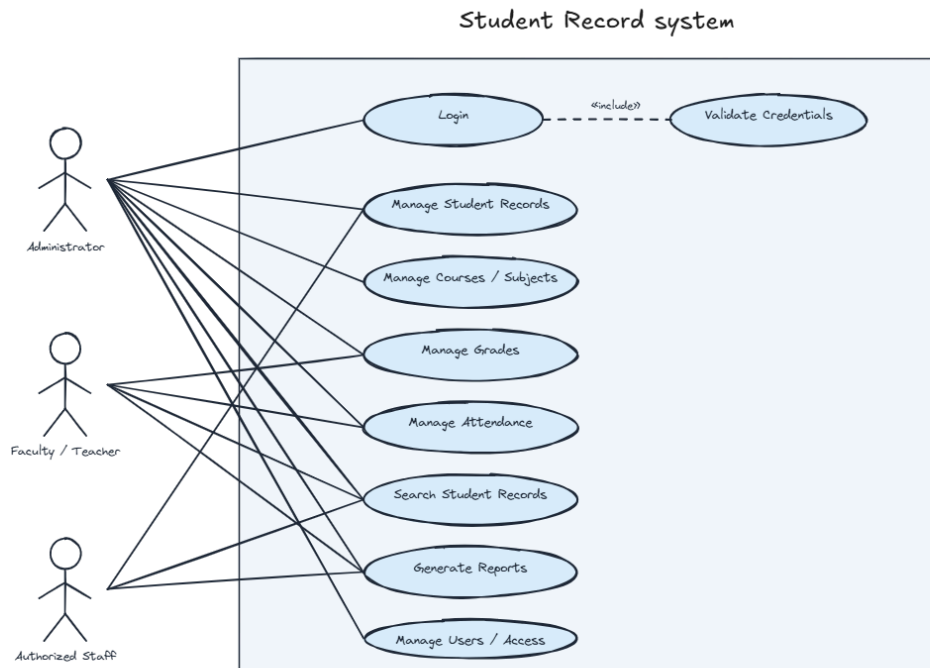
TC-[MODULE]-[NUMBER]

Examples:

- TC-STD-01 – Student Management
- TC-SUB-01 – Subject Management
- TC-GRD-01 – Grade Management
- TC-ATT-01 – Attendance Management
- TC-SRCH-01 – Record Search
- TC-RPT-01 – Reporting
- TC-AUTH-01 – Authentication
- TC-DATA-01 – Data Management
- TC-ADMIN-01 – Administration
- TC-EXIT-01 – System Exit

SECTION 7

7. System Models and Diagrams :



SECTION 8

8. Requirements Traceability Matrix (RTM)

Student Record System — Team 11

Status legend: N = Not Tested, P = Passed, A = Action Required

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
SRS-F-001	Add new student record	4.1 / DS-STU-01	Student Management Module	TC-STD-01	N	Valid student record is successfully stored
SRS-F-002	View student information	4.1 / DS-STU-02	Student Management Module	TC-STD-02	N	Selected student's permitted information is displayed correctly
SRS-F-003	Update student information	4.1 / DS-STU-03	Student Management Module	TC-STD-03	N	Updated student information is saved and displayed correctly
SRS-F-004	Prevent duplicate student IDs	4.1 / DS-STU-04	Student Management Module	TC-STD-04	N	Duplicate student ID attempt is rejected
SRS-F-005	Add and maintain subject records	4.2 / DS-SUB-01	Course / Subject Management Module	TC-SUB-01	N	Valid subject record is created and stored
SRS-F-006	View available subjects	4.2 / DS-SUB-02	Course / Subject Management Module	TC-SUB-02	N	Subject details are displayed correctly
SRS-F-007	Assign subjects to students	4.2 / DS-SUB-03	Course / Subject Management Module	TC-SUB-03	N	Assigned subject appears in student's academic record
SRS-F-008	Enter student grades	4.3 / DS-GRD-01	Grade Management Module	TC-GRD-01	N	Valid grade is stored against correct student and subject
SRS-F-009	Validate grades	4.3 / DS-GRD-02	Grade Management Module	TC-GRD-02	N	Invalid grade values are rejected
SRS-F-010	Update student grades	4.3 / DS-GRD-03	Grade Management Module	TC-GRD-03	N	Modified grade is correctly stored and displayed
SRS-F-011	Calculate GPA and CGPA	4.3 / DS-GRD-04	Grade Management Module	TC-GRD-04	N	Calculated GPA/CGPA matches expected result
SRS-F-012	Record student attendance	4.4 / DS-ATT-01	Attendance Management Module	TC-ATT-01	N	Attendance is recorded for correct student and subject
SRS-F-013	Update attendance records	4.4 / DS-ATT-02	Attendance Management Module	TC-ATT-02	N	Updated attendance information is correctly stored
SRS-F-014	Calculate attendance percentage	4.4 / DS-ATT-03	Attendance Management Module	TC-ATT-03	N	Displayed percentage matches recorded attendance

SRS-F-015	Search for students	4.5 / DS-SRCH-01	Record Search and Reporting Module	TC-SRCH-01	N	Matching student records are displayed
SRS-F-016	Generate student academic reports	4.5 / DS-RPT-01	Record Search and Reporting Module	TC-RPT-01	N	Generated report contains student's correct records
SRS-F-017	View grade sheets and attendance summaries	4.5 / DS-RPT-02	Record Search and Reporting Module	TC-RPT-02	N	Selected grade sheet or attendance summary is displayed correctly
SRS-F-018	Authenticate users	4.6 / DS-AUTH-01	User / Access Management Module	TC-AUTH-01	N	Valid credentials provide access and invalid credentials are rejected
SRS-F-019	Provide role-based access	4.6 / DS-AUTH-02	User / Access Management Module	TC-AUTH-02	N	Users can access only functions permitted for their role
SRS-F-020	Prevent students from modifying protected records	4.6 / DS-AUTH-03	User / Access Management Module	TC-AUTH-03	N	Unauthorized modification attempts are rejected
SRS-F-021	Store records in persistent storage	4.7 / DS-DATA-01	Data Management Module	TC-DATA-01	N	Stored information remains available after restarting the system
SRS-F-022	Validate required information	4.7 / DS-DATA-02	Data Management Module	TC-DATA-02	N	Records with missing required fields are rejected
SRS-F-023	Maintain record consistency	4.7 / DS-DATA-03	Data Management Module	TC-DATA-03	N	Invalid references cannot be stored
SRS-F-024	Provide logout or exit	4.8 / DS-EXIT-01	System Exit / Administration Module	TC-EXIT-01	N	Logout or exit safely terminates the active session
SRS-F-025	Manage authorized user accounts	4.8 / DS-ADMIN-01	System Exit / Administration Module	TC-ADMIN-01	N	Administrators can create, modify, or deactivate permitted accounts
SRS-N F-001	System response time	5.1 / DS-PERF-01	Entire System	TC-Perf-01	N	95th percentile response time ≤ 3s in load testing
SRS-N F-002	Reliable operation	5.1 / DS-REL-01	Entire System	TC-REL-01	N	System completes normal operations without unexpected termination or data loss.
SRS-N F-003	Protect stored student personal data	5.1 / DS-SEC-01	Security / Data Management	TC-Sec-01	N	Encryption audit passes; no plaintext PII in logs or database
SRS-N F-004	Maintain CRUD audit logs	5.1 / DS-AUDIT-01	Audit / Data Retention	TC-OPS-01	N	Automated logging and archival are verified

SRS-N F-005	Support accessibility features	5.1 / DS-UX-01	User Interface Module	TC-UX-01	N	Accessibility audit passes
SRS-N F-006	Handle expected student records efficiently	5.1 / DS-SCALE-01	Entire System	TC-Perf-02	N	System successfully processes the defined test dataset without significant performance degradation during normal operation.
SRS-N F-007	Provide role-based access control	5.1 / DS-RBAC-01	Security / Access Control	TC-Sec-02	N	RBAC matrix verified; unauthorized access attempts are blocked
SRS-S R-001	Protect stored student information	5.3 / DS-SEC-01	Security / Data Management	TC-Sec-01	N	Unauthorized attempts to access protected student information are rejected.
SRS-S R-002	Use secure password hashing	5.3 / DS-SEC-02	Authentication Module	TC-Sec-02	N	Passwords stored as hashes; plaintext not found in logs/database
SRS-S R-003	Enforce 15-minute session timeout	5.3 / DS-SEC-03	Session Management Module	TC-Sec-03	N	Session terminates at 15 minutes and user must re-authenticate
SRS-S R-004	Log failed login attempts and lock after 5 failures	5.3 / DS-SEC-04	Authentication / Audit Module	TC-Sec-04	N	Failed attempts are logged; account locks after 5 failures
SRS-S R-005	Enforce role-based access restrictions	5.3 / DS-SEC-05	Access Control Module	TC-Sec-05	N	RBAC enforcement verified for all roles; unauthorized actions blocked

Coverage: 25 Functional Requirements (SRS-F-001–025), 7 Non-Functional Requirements (SRS-NF-001–007), and 5 Security Requirements (SRS-SR-001–005).